The Global Landscape and Clinical Translation of Personalized Neoantigen Vaccines in Non-Small Cell Lung Cancer: A Scoping Review Protocol

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Abstract

Background: Non-small cell lung cancer (NSCLC) remains a leading cause of cancer-related mortality worldwide despite advances in targeted therapies and immune checkpoint inhibitors. Therapeutic cancer vaccines have evolved from conventional approaches targeting shared tumor-associated antigens toward personalized strategies based on patient-specific somatic mutations. Advances in next-generation sequencing, computational immunology, and vaccine manufacturing have facilitated the development of personalized neoantigen vaccines. However, the historical evolution, global clinical trial landscape, and consistency of international trial registration information require systematic mapping to understand the clinical translation of these therapeutic strategies.

Objective: This scoping review aims to examine the historical development of therapeutic vaccines for NSCLC, characterize the global clinical trial landscape of personalized neoantigen vaccines, and assess the coverage and consistency of clinical trial information between ClinicalTrials.gov and the World Health Organization International Clinical Trials Registry Platform (WHO ICTRP).

Methods: The review will follow the Joanna Briggs Institute (JBI) methodology for scoping reviews and will be reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR). A multi-stage search strategy will be employed. A historical search will identify therapeutic NSCLC vaccine trials registered in ClinicalTrials.gov from database inception to 17 August 2026. A focused search will identify personalized neoantigen vaccine trials from ClinicalTrials.gov and WHO ICTRP using the original predefined Boolean search expression. A supplementary targeted search using investigational product names and alternative intervention terminology will be conducted to identify eligible trials potentially missed by the original search. All included trials must have been registered on or before 17 August 2026. Two independent reviewers will screen records against predefined eligibility criteria. Data will be charted using a standardized extraction form. Overlapping registry records will be linked to identify unique trials, while independent clinical protocols evaluating the same vaccine product

will be counted separately. Descriptive analyses and visualizations will be performed using R software.

Expected contribution: This review will provide a structured evidence map of therapeutic NSCLC vaccine development, the global clinical translation of personalized neoantigen vaccine platforms, and the consistency of information reported across international clinical trial registries.

Keywords: Non-small cell lung cancer; personalized neoantigen vaccines; therapeutic cancer vaccines; precision oncology; clinical trials; clinical translation; scoping review; ClinicalTrials.gov; WHO ICTRP.

1. Introduction

1.1 Background

Non-small cell lung cancer (NSCLC) accounts for approximately 85% of lung cancer diagnoses and remains a major cause of cancer-related morbidity and mortality worldwide[1]. Although there have been improvements in NSCLC treatment, tumor recurrence has remained a major obstacle. Immune checkpoint inhibitors, individual-specific targeted therapies, and multimodal treatment approaches are examples of treatment advancements that have significantly improved outcomes in certain patient populations. However, despite these improvements, notable challenges like NSCLC treatment resistance, tumor diversity, and disease recurrence still exist. Therapeutic cancer vaccines represent a category of active immunotherapy aimed at eliciting an immune response that targets antigens present on cancer cells. Commonly researched tumor-associated antigens in therapeutic vaccine research for NSCLC include epidermal growth factor (EGF), mucin 1 (MUC1), carcinoembryonic antigen (CEA), melanoma-associated antigens (MAGE), and NY-ESO-1. To develop these vaccines, several strategies have been utilized, including synthetic peptides, recombinant proteins, whole-tumor cells, dendritic cells, and viral or bacterial vectors, but these are limited since the approach is not specific to the tumor.

To overcome this limitation, advances using personalized neoantigen vaccines are being encouraged. These vaccines use tumor-specific peptides (neoantigens) produced by somatic mutations and displayed by major histocompatibility complex (MHC) molecules. Antigen-specific CD4+ and CD8+ T cells help in the identification of these peptides, leading to anticancer immune responses. They are created using mutations found in a patient's tumor, in contrast to traditional immunizations that contain predefined common antigens in populations. The steps encompassing the development process include Tumor genomic and transcriptome analysis, somatic variant discovery, human leukocyte antigen (HLA) genotyping, computational neoepitope prediction, candidate neoantigen prioritization, and customized vaccine production [2–7].

The advances in next-generation sequencing, computational immunology, and bioinformatics have also improved identification of candidate neoantigens and the development of personalized vaccine

platforms, including synthetic peptide vaccines, RNA/mRNA vaccines, dendritic-cell-based vaccines, DNA vaccines, and viral-vector vaccines. These personalized neoantigen vaccines could be evaluated as either monotherapy or combined with immune checkpoint inhibitors, chemotherapy, radiotherapy, or molecularly targeted therapies. Their clinical translation depends on several factors, including vaccine manufacturing feasibility, treatment timing, patient selection, immunogenicity assessment, clinical trial design, and evaluation of meaningful clinical endpoints.

Clinical trial registries such as ClinicalTrials.gov and WHO ICTRP provide important information for characterizing these developments. They contain information on which clinical trials have been registered. They also contain details of the clinical trials such as clinical trial phases, vaccine interventions administered to patients, recruitment status of the trial, planned enrollment, sponsorship of the trial, geographical distribution of the study sites, and clinical outcomes expected from the trial. However, differences in registry coverage, reporting terminology, indexing practices, and record update frequency may affect the completeness and consistency of available clinical trial information in the database/ registries. Another limitation is that some investigational neoantigen vaccines may also be registered under proprietary product names, investigational codes, or regulatory drug nomenclature rather than generic vaccine terminology.

Mapping of registered clinical trials systematically, while being supported by supplementary targeted searches, is therefore necessary to characterize the historical development and global clinical translation of personalized neoantigen vaccines in NSCLC.

1.2 Rationale for the review

To better understand the clinical translation of personalized neoantigen vaccines, an assessment of both historical therapeutic vaccine development and the characteristics of contemporary clinical investigations is necessary. ClinicalTrials.gov registry, which has registered clinical trials, was launched in February 2000 and provides a defined starting point for systematic registration-based analyses. Some studies were initiated before 2000 and thus were registered retrospectively in this database. As a result, the trial registration dates and actual study initiation dates are different and will be distinguished when examining historical trends of NSCLC clinical trials. The review will also characterize clinical development phases, vaccine platforms, recruitment status, planned enrollment, sponsorship, geographical distribution, therapeutic combinations, and clinical endpoints. The study will also include a supplementary targeted search to improve identification of eligible personalized neoantigen vaccine trials that may not be retrieved through the original Boolean search expression. Because the targeted search may identify trials registered before the original search cutoff, these records will be assessed using the same eligibility criteria and registration period as the primary search. Besides trends, the study will make a comparison of ClinicalTrials.gov and WHO ICTRP records to identify overlapping registrations and differences in reporting completeness and recruitment status. This will be possible through a scoping review, appropriate

for its principal purpose of systematically identifying, classifying, and mapping the extent and characteristics of registered clinical research rather than synthesizing pooled estimates of therapeutic effectiveness.

1.3 Review objectives

The overall objective will be to systematically map the historical development of therapeutic NSCLC vaccines and characterize the global clinical translation of personalized neoantigen vaccines using international clinical trial registry data.

Specific objectives

1. To examine the historical development of therapeutic NSCLC vaccine trials by comparing temporal trends in conventional shared-antigen vaccines and personalized neoantigen vaccines using trial registration and study initiation dates.
2. To characterize the global clinical trial landscape of personalized neoantigen vaccines in NSCLC, including vaccine platforms, clinical phases, recruitment status, planned enrollment, sponsorship, geographical distribution, therapeutic combinations, and clinical outcome measures.
3. To assess the overlap, coverage, and consistency of personalized neoantigen vaccine trial registrations for NSCLC between ClinicalTrials.gov and WHO ICTRP, with particular emphasis on recruitment-status discrepancies and differences in reported trial characteristics.

1.4 Review questions

The review will address the following questions:

1. How have therapeutic NSCLC vaccine trials evolved, particularly between conventional shared-antigen and personalized neoantigen vaccine strategies?
2. What are the characteristics and geographical distribution of registered clinical trials investigating personalized neoantigen vaccines in NSCLC?
3. What differences exist in the coverage, completeness, and consistency of personalized neoantigen vaccine trial information between ClinicalTrials.gov and WHO ICTRP?

2. Materials and Methods

2.1 Study design and reporting framework

This study will be conducted as a scoping review following the Joanna Briggs Institute methodological framework. The completed review will be reported in accordance with the PRISMA-ScR checklist [8,9]. The search will be conducted by two independent reviewers in August 2026.

Relevant PRISMA 2020 recommendations will additionally inform documentation of records identified through supplementary search methods. The review will comprise a historical therapeutic NSCLC vaccine analysis and a focused personalized neoantigen vaccine analysis. A multi-stage search approach will be employed, incorporating A Primary systematic registry search, and a Supplementary targeted clinical development search.

The original and supplementary search pathways will be reported independently before integrating eligible records.

2.2 Eligibility criteria

Eligibility criteria will be developed using the Population–Concept–Context (PCC) framework.

2.2.1 Population

The population will include patients diagnosed with NSCLC, including adenocarcinoma, squamous cell carcinoma, large-cell carcinoma, and unspecified NSCLC. Trials enrolling multiple solid-tumor types will be eligible when NSCLC is explicitly included in the study population or eligibility criteria. Trials exclusively involving small-cell lung cancer or other thoracic malignancies without an eligible NSCLC population will be excluded.

2.2.2 Concept

The historical analysis will include active therapeutic vaccines designed to stimulate antigen-specific immune responses in NSCLC. Eligible vaccines will be classified into two categories, namely: 1) the conventional shared-antigen vaccines: Non-personalized therapeutic vaccines targeting shared tumor-associated antigens or broader tumor antigen preparations. 2) Personalized neoantigen vaccines: Therapeutic vaccines developed using patient-specific tumor somatic mutations and individualized neoantigen selection. The focused clinical trial landscape and cross-registry analyses will include only personalized neoantigen vaccines.

2.2.3 Context

The review will include registered interventional clinical trials conducted globally without geographical restrictions. The historical analysis will use ClinicalTrials.gov records, including retrospectively registered trials with study initiation dates preceding the establishment of the registry. The focused personalized neoantigen vaccine analysis will use ClinicalTrials.gov and WHO ICTRP [10,11]. The registration cutoff for all analyses will be August 2026.

2.2.4 Inclusion criteria

The focused personalized neoantigen vaccine analysis will include:

1. Registered interventional clinical trials investigating therapeutic personalized neoantigen vaccines.
2. Trials enrolling patients with NSCLC, either exclusively or within broader solid-tumor populations.
3. Trials involving patient-specific neoantigen selection derived from individual tumor mutation profiles.
4. Trials investigating peptide, RNA/mRNA, DNA, viral-vector, dendritic-cell, or other personalized neoantigen vaccine platforms.
5. Trials evaluating personalized neoantigen vaccines alone or in combination with other anticancer therapies.
6. Trials first registered on or before 17 August 2026, regardless of clinical phase or recruitment status.

2.2.5 Exclusion criteria

The following will be excluded:

1. Observational or non-interventional studies without administration of a therapeutic vaccine.
2. Preclinical and animal investigations.
3. Trials exclusively involve small cell lung cancer.
4. Studies without an eligible NSCLC population.
5. Prophylactic vaccines targeting infectious diseases.
6. Passive antibody therapies without an active therapeutic vaccine component.
7. Adoptive cellular therapies without administration of an identifiable neoantigen vaccine.
8. Vaccines targeting only shared non-mutated tumor-associated antigens, for the focused neoantigen analysis.
9. Fixed, non-personalized neoantigen vaccines without individualized antigen selection.
10. Records in which vaccine-related terminology appears only in background text or exclusion criteria.
11. Records with insufficient information to establish eligibility.
12. Trials first registered after August 2026.

Conventional therapeutic NSCLC vaccines will remain eligible for the historical analysis.

2.3 Information sources

The primary clinical trial registries will be **ClinicalTrials.gov (**https://clinicaltrials.gov/) and the **WHO International Clinical Trials Registry Platform (**https://trialsearch.who.int/). The WHO ICTRP provides access to trial records from participating international registries.

Supplementary sources may include publicly available regulatory documents, sponsor disclosures, clinical development program descriptions, and linked trial publications. Trials identified through supplementary sources will be verified against available trial registry records before inclusion.

2.4 Search strategy

Two complementary search pathways will be used.

2.4.1 Historical therapeutic NSCLC vaccine search

A historical search of ClinicalTrials.gov will be undertaken to identify registered interventional therapeutic NSCLC vaccine trials. The search will cover records from database inception through August 2026. Trials initiated before the establishment of ClinicalTrials.gov will remain eligible if they were subsequently registered on or before the cutoff. The historical search will use NSCLC-related and therapeutic vaccine-related search terms. Eligible records will be classified into conventional shared-antigen and personalized neoantigen vaccine categories. The historical analysis will use registration dates and study initiation dates as distinct temporal variables. The complete executable search strategy, including applied filters, will be documented.

2.4.2 Primary personalized neoantigen vaccine search

The primary search will retain the original expression used in the initial review:

(("Non-small cell lung cancer" OR "NSCLC") AND ("neoantigen" OR "neo-epitope" OR "personalized vaccine")). This expression will be applied to ClinicalTrials.gov and WHO ICTRP. Registry-specific technical adaptations will be documented while preserving the original search concepts. Eligible trials must have been registered on or before August 2026.

2.4.3 Supplementary targeted search

A supplementary targeted search will be conducted to identify eligible personalized neoantigen vaccine trials potentially missed by the original registry search. This search will use investigational product names, vaccine development codes, and alternative terminology.The supplementary expression will include:

("Non-Small Cell Lung Cancer" OR "NSCLC")AND("mRNA-4157"OR "V940"OR "intismeran autogene"OR "INTerpath"OR "Individualized Neoantigen Therapy"). The supplementary search will be restricted to trials first registered on or before 17 August 2026. Trials identified through this

pathway will undergo the same eligibility screening, data extraction, and deduplication procedures as records retrieved through the primary search. The supplementary search is intended to improve completeness within the original review period rather than extend the registration cutoff.

2.4.4 Search dates and registry data snapshots

The registration cutoff and date of information retrieval will be treated as separate variables. The registration cutoff will remain August 2026. The date on which the supplementary search is conducted will be recorded explicitly. For each included record, the date of the registry's most recent update at extraction will be recorded where available. If trial information has changed after 17 August 2026, the analysis will distinguish historical information available by the cutoff from subsequently updated registry data whenever historical versions are accessible. When historical record versions cannot be retrieved, the information will be identified as reflecting the later extraction date rather than being represented as an August 2026 snapshot. This distinction will be considered particularly important when interpreting recruitment status, planned enrollment, study completion dates, and cross-registry discrepancies.

2.4.5 Search documentation

For each search pathway, the following will be recorded: 1) Database or source;2)Search execution date; 3) Registration cutoff; 4) Exact search expression; 5) Search fields and filters; 6) Retrieved registration identifiers; 7) Eligibility decisions; and 8)Duplicate and overlapping registration assessment.

2.5 Study selection

Two independent reviewers will screen retrieved records against predefined eligibility criteria. Screening will involve trial titles, intervention descriptions, eligibility information, and relevant registry metadata. Disagreements will be resolved through discussion and, where necessary, consultation with a third reviewer. Reasons for exclusion will be documented. Primary and supplementary search records will initially be retained separately. Eligible records will then be linked and deduplicated to establish the unique clinical trial dataset. The selection process will be reported using a PRISMA-ScR flow diagram.

2.6 Clinical trial identification and accounting

The primary unit of analysis will be the unique registered clinical trial protocol rather than the investigational vaccine product. Each independent clinical trial protocol will be counted separately. Trials evaluating the same personalized neoantigen vaccine product under different clinical protocols will be retained as distinct studies. For example, INTerpath-002 (NCT06077760) and INTerpath-009 (NCT06623422) will be evaluated as separate trial protocols investigating the same personalized mRNA neoantigen vaccine product, subject to eligibility confirmation. On the other

hand, registrations referring to the same clinical protocol across ClinicalTrials.gov and WHO ICTRP will be linked and counted once in the integrated dataset. Trial matching will use primary registration identifiers, secondary identifiers, official titles, sponsor information, intervention characteristics, and relevant study dates. Separate variables will be maintained for: 1)Unique Clinical trial identifier; 2) Registry-specific registration identifier; 3) Vaccine product identifier; 4) Vaccine platform; 5) Trial protocol or development program. This approach will distinguish the number of clinical trials from the number of unique investigational vaccine products.

2.7 Classification of therapeutic vaccine modalities

Therapeutic NSCLC vaccines will be classified into: 1) **Conventional shared-antigen vaccines:** Non-personalized therapeutic vaccines targeting shared tumor-associated antigens or broader tumor antigen preparations; 2) **Personalized neoantigen vaccines:** Vaccine formulations developed through patient-specific tumor mutation identification and individualized neoantigen selection. The focused personalized neoantigen analysis will include peptide, RNA/mRNA, DNA, viral-vector, dendritic-cell, and other individualized vaccine delivery technologies. Conventional vaccines will be retained only for the historical analysis.

2.8 Data extraction and charting

A standardized extraction template will be developed and piloted. Two reviewers will independently extract or verify relevant trial information.

Domain	Variables
Trial identification	Trial ID, title, registry source, secondary identifiers
Vaccine identity	Product name, investigational code, platform
Chronology	Registration date, first-posted date, study start date
Study design	Clinical phase, allocation, masking, intervention model
Population	NSCLC subtype, clinical stage, treatment setting
Recruitment	Status and relevant trial dates
Enrollment	Planned or actual enrollment
Sponsorship	Industry, academic, hospital, or other
Geographical distribution	Sponsor country, participating countries, study sites

Therapeutic combinations	Checkpoint inhibitors, chemotherapy, radiotherapy, targeted therapy, monotherapy
Primary endpoints	Safety, tolerability, immunogenicity, clinical efficacy
Secondary endpoints	Immune responses, ORR, PFS, DFS, OS
Registry reporting	Record update dates, missing information, discrepancies

Missing information will be coded as "Not reported." Original registry terminology will be retained alongside standardized categories. For mixed solid-tumor trials, overall enrollment and NSCLC-specific enrollment will be distinguished where reported.

2.9 Classification of clinical development phases

Clinical trial phases will be extracted from the registry records and classified as: 1) Early Phase I; 2) Phase I; 3) Phase I/II;4) Phase II ;5) Phase II/III; 6) Phase III; Phase IV, if applicable; Not applicable or not reported. Mixed-phase studies will remain separately classified.

2.10 Classification of therapeutic combinations

Treatment combinations will be classified as: 1) Vaccine monotherapy; 2) Vaccine plus immune checkpoint inhibitor; 3) Vaccine plus chemotherapy; 4) Vaccine plus radiotherapy; 5) Vaccine plus targeted therapy; 6) Other or multimodal combinations.

Trials evaluating multiple concomitant treatments will be coded in all applicable categories.

2.11 Classification of clinical outcome measures

Clinical outcome measures will be categorized into: 1) **Safety and tolerability:** Adverse events, serious adverse events, dose-limiting toxicities, maximum tolerated dose, and other safety assessments; 2) **Immunogenicity:** Neoantigen-specific T-cell responses, CD4+ and CD8+ T-cell activation, cytokine responses, and other immune assays; 3) **Clinical efficacy:** Objective response rate, disease control rate, progression-free survival, disease-free survival, overall survival, and other clinical outcomes. Registered efficacy endpoints will be distinguished from demonstrated treatment effectiveness.

2.12Cross-registry comparison

ClinicalTrials.gov and WHO ICTRP records referring to the same clinical trial will be matched. The Comparisons will assess: 1) Recruitment status; 2) Clinical phase; 3) Planned enrollment; 4) Study start and completion dates; 5) Sponsor information; 6) Intervention descriptions; 7) Registry update dates; 8) Missing or incomplete data.

Recruitment-status differences will be classified as: 1) **Potential update delay:** Differences potentially reflecting outdated or incomplete status information; 2) **Substantive discrepancy:** Differences suggesting materially inconsistent descriptions of trial activity ;3) **Terminology difference:** Differences attributable to registry-specific status definitions or conventions; 4) **Concordance:** Equivalent recruitment information across corresponding records.

Where possible, record update dates will be considered when interpreting discrepancies.

Registry data obtained after August 2026 will not be assumed to represent information available on the original cutoff date.

2.13 Statistical analysis

Data will be managed and analyzed using R statistical software, with tidyverse for data transformation and ggplot2 for visualizations. Categorical variables will be summarized using frequencies and percentages. Continuous variables will be summarized using appropriate descriptive statistics.

2.13.1 Objective 1: Historical development

Annual and cumulative trends in therapeutic NSCLC vaccine trials will be analyzed according to conventional shared-antigen and personalized neoantigen vaccine categories. First-posted registration dates will be used for registration trends. Study start dates will be analyzed separately to characterize historical initiation patterns, including retrospectively registered trials initiated before 2000.

2.13.2 Objective 2: Global clinical trial landscape

Personalized neoantigen vaccine trials will be summarized according to clinical phase, vaccine platform, recruitment status, planned enrollment, sponsor category, geographical distribution, therapeutic combinations, and endpoints. Multinational trials will contribute to each participating country's count but will be counted once in global unique-trial totals. The number of distinct vaccine products will be described separately from the number of clinical trials where sufficient information is available.

2.13.3 Objective 3: Cross-registry comparison

Registry overlap will be assessed using linked clinical trial identifiers. Frequencies and percentages of concordant and discrepant records will be calculated. The interpretation of registry discrepancies will account for differences in record update dates and data availability. No formal meta-analysis of therapeutic efficacy will be conducted.

2.14 Presentation of findings

Findings will be presented according to the three review objectives.

- **Objective 1:** Historical annual and cumulative trends in conventional and personalized therapeutic NSCLC vaccines.
- **Objective 2:** Global clinical trial characteristics, vaccine platforms, clinical phases, geographical distribution, therapeutic combinations, and endpoints.
- **Objective 3:** Registry coverage, overlap, concordance, and discrepancies.

A PRISMA-ScR flow diagram will document the identification and selection of eligible trial records.

2.15 Critical appraisal

Formal risk-of-bias assessment will not be performed because this scoping review aims to map registered clinical research rather than establish intervention effectiveness. Registry completeness and reporting consistency will be assessed descriptively.

2.16 Ethical considerations

This review will use publicly accessible clinical trial registry records. No direct patient recruitment, clinical intervention, or collection of identifiable patient-level information will be undertaken. Formal ethical approval is therefore not anticipated to be required, subject to institutional policies.

2.17 Protocol registration and amendments

The original review manuscript identifies an existing Open Science Framework. The registered protocol and amendment history will be reviewed before submission. The supplementary targeted search and any methodological changes will be documented transparently. The registration cutoff will remain August 2026, even where supplementary searches are executed after this date. The original search and supplementary trial identification procedures will be reported separately.

3. Anticipated Limitations

The review may be limited by incomplete or outdated clinical trial registry records, differences in indexing terminology, and inconsistent reporting practices. The retrospective registration of older clinical trials may complicate comparisons between trial registration dates and actual initiation dates. The original focused search strategy may fail to retrieve eligible personalized neoantigen vaccine trials registered under proprietary product names or alternative terminology. Although the supplementary targeted search is intended to improve identification, it may not eliminate all limitations relating to registry coverage.

Additionally, trial records updated after the registration cutoff may not fully reflect the information available at the original search date, particularly when historical registry versions are unavailable. Mixed solid-tumor trials may not provide NSCLC-specific enrollment or clinical outcome information. The review will characterize the global clinical research landscape rather than establish the clinical effectiveness of personalized neoantigen vaccines.

4. Dissemination

Findings will be disseminated through publication in a peer-reviewed scientific journal and presentations at relevant oncology, immunology, and precision medicine conferences. Where appropriate, search strategies, data extraction templates, and non-confidential trial-level datasets will be shared in an open research repository to support transparency and reproducibility.

5. Conclusion

This protocol outlines a systematic approach to mapping the historical development and global clinical translation of personalized neoantigen vaccines in non-small cell lung cancer. The review will combine an assessment of therapeutic NSCLC vaccine development, a detailed characterization of personalized neoantigen vaccine trials, and a comparison of international clinical trial registry information. By maintaining a consistent registration cutoff of 17 August 2026 and documenting supplementary targeted searches separately, the review aims to provide a transparent and reproducible account of the clinical development landscape. The findings are intended to inform future clinical trial design, registry reporting standardization, and international research collaboration in personalized neoantigen-based immunotherapy.

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